

The LeetCode Pattern Playbook

Your 27 patterns, plus 13 more that round out an interview-ready set. Each group teaches one reusable technique. Check problems off, search by name or number, and track your progress. Progress saves automatically.

0 / 208 solved · 0%

Search by problem name, number, or pattern...

40 patterns · 208 problems

- All
- Easy
- Medium
- Hard
- Unsolved only

PATTERN 01

Sliding Window

5 MED 1 HARD

Grow/shrink a window; track a running condition instead of re-scanning.

- ☐ 3 Longest Substring Without Repeating Characters M
- ☐ 76 Minimum Window Substring H
- ☐ 209 Minimum Size Subarray Sum M
- ☐ 424 Longest Repeating Character Replacement M
- ☐ 567 Permutation in String M
- ☐ 904 Fruit Into Baskets M

PATTERN 02

Two Pointers

5 MED 1 HARD

Converging or same-direction pointers on sorted data to cut $O(n^2)$ to $O(n)$.

- ☐ 11 Container With Most Water M
- ☐ 15 3Sum M
- ☐ 16 3Sum Closest M
- ☐ 18 4Sum M
- ☐ 42 Trapping Rain Water H
- ☐ 167 Two Sum II - Input Array Is Sorted M

PATTERN 03

PATTERN 04

Fast / Slow Pointers

4 EASY

2 MED

Tortoise-and-hare for cycle detection, midpoints, and nth-from-end.

- ☐ 141 Linked List Cycle E
- ☐ 142 Linked List Cycle II M
- ☐ 19 Remove Nth Node From End of List M
- ☐ 876 Middle of the Linked List E
- ☐ 160 Intersection of Two Linked Lists E
- ☐ 234 Palindrome Linked List E

Binary Search on Sorted Data

2 EASY

4 MED

Halve the search space; watch boundary and rotation invariants.

- ☐ 33 Search in Rotated Sorted Array M
- ☐ 34 Find First and Last Position of Element in Sorted Array M
- ☐ 35 Search Insert Position E
- ☐ 153 Find Minimum in Rotated Sorted Array M
- ☐ 162 Find Peak Element M
- ☐ 704 Binary Search E

PATTERN 05

Binary Search on Answer

4 MED

2 HARD

Binary search the result value; use a feasibility predicate.

- ☐ 875 Koko Eating Bananas M
- ☐ 1011 Capacity To Ship Packages Within D Days M
- ☐ 410 Split Array Largest Sum H
- ☐ 774 Minimize Max Distance to Gas Station H
- ☐ 1283 Find the Smallest Divisor Given a Threshold M
- ☐ 1482 Minimum Number of Days to Make m Bouquets M

PATTERN 06

Hashing / Frequency Maps

3 EASY

3 MED

Trade space for $O(1)$ lookups; count, group, or dedupe.

- ☐ 1 Two Sum E
- ☐ 49 Group Anagrams M
- ☐ 128 Longest Consecutive Sequence M
- ☐ 217 Contains Duplicate E
- ☐ 242 Valid Anagram E
- ☐ 347 Top K Frequent Elements M

PATTERN 07

PATTERN 08

Prefix Sum / Running Sum

2 EASY

4 MED

Precompute cumulative sums; answer range queries in $O(1)$.

- ☐ 303 Range Sum Query - Immutable E
- ☐ 560 Subarray Sum Equals K M
- ☐ 724 Find Pivot Index E
- ☐ 930 Binary Subarrays With Sum M
- ☐ 974 Subarray Sums Divisible by K M

Difference Array / Range Updates

1 EASY

5 MED

Mark deltas at boundaries; one pass reconstructs all range updates.

- ☐ 370 Range Addition M
- ☐ 1094 Car Pooling M
- ☐ 1109 Corporate Flight Bookings M
- ☐ 1893 Check if All the Integers in a Range Are Covered E
- ☐ 1943 Describe the Painting M

PATTERN 09

Monotonic Stack

1 EASY

3 MED

2 HARD

Maintain increasing/decreasing stack for next-greater/span problems.

- ☐ 739 Daily Temperatures M
- ☐ 496 Next Greater Element I E
- ☐ 503 Next Greater Element II M
- ☐ 84 Largest Rectangle in Histogram H
- ☐ 85 Maximal Rectangle H
- ☐ 901 Online Stock Span M

PATTERN 10

Monotonic Queue / Deque

2 MED

4 HARD

Deque keeps window extremes in $O(1)$ amortized.

- ☐ 239 Sliding Window Maximum H
- ☐ 862 Shortest Subarray with Sum at Least K H
- ☐ 1425 Constrained Subsequence Sum H
- ☐ 1438 Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit M
- ☐ 1499 Max Value of Equation H
- ☐ 1696 Jump Game VI M

PATTERN 11

Heap / Top K

2 EASY

4 MED

PATTERN 12

Intervals

1 EASY

5 MED

Heap of size k for streaming top/closest/ k th queries.

- ☐ 215 Kth Largest Element in an Array M
- ☐ 347 Top K Frequent Elements M
- ☐ 692 Top K Frequent Words M
- ☐ 703 Kth Largest Element in a Stream E
- ☐ 973 K Closest Points to Origin M

Sort by start/end; merge, insert, or sweep for overlaps.

- ☐ 56 Merge Intervals M
- ☐ 57 Insert Interval M
- ☐ 252 Meeting Rooms E
- ☐ 253 Meeting Rooms II M
- ☐ 435 Non-overlapping Intervals M
- ☐ 452 Minimum Number of Arrows to Burst Balloons M

PATTERN 13

Greedy Scheduling / Sorting

6 MED

Make the locally optimal choice; prove it stays globally optimal.

- ☐ 45 Jump Game II M
- ☐ 55 Jump Game M
- ☐ 406 Queue Reconstruction by Height M
- ☐ 621 Task Scheduler M
- ☐ 763 Partition Labels M
- ☐ 134 Gas Station M

PATTERN 14

Linked List Manipulation

1 EASY

3 MED

2 HARD

Pointer rewiring, dummy heads, and in-place reversal.

- ☐ 21 Merge Two Sorted Lists E
- ☐ 23 Merge k Sorted Lists H
- ☐ 24 Swap Nodes in Pairs M
- ☐ 25 Reverse Nodes in k-Group H
- ☐ 92 Reverse Linked List II M
- ☐ 138 Copy List with Random Pointer M

PATTERN 15

Tree DFS

4 EASY

1 MED

1 HARD

Recurse and combine child results; classic post-order aggregation.

- ☐ 104 Maximum Depth of Binary Tree E
- ☐ 112 Path Sum E

PATTERN 16

Tree BFS / Level Order

1 EASY

5 MED

Queue by level; snapshot the size before each layer.

- ☐ 102 Binary Tree Level Order Traversal M

- ☐ 113 Path Sum II M
- ☐ 543 Diameter of Binary Tree E
- ☐ 124 Binary Tree Maximum Path Sum H
- ☐ 226 Invert Binary Tree E

- ☐ 103 Binary Tree Zigzag Level Order Traversal M
- ☐ 199 Binary Tree Right Side View M
- ☐ 515 Find Largest Value in Each Tree Row M
- ☐ 637 Average of Levels in Binary Tree E
- ☐ 116 Populating Next Right M

PATTERN 17

BST Problems

1 EASY 5 MED

Exploit the sorted in-order property for search and validation.

- ☐ 98 Validate Binary Search Tree M
- ☐ 99 Recover Binary Search Tree M
- ☐ 230 Kth Smallest Element in a BST M
- ☐ 235 Lowest Common Ancestor of a Binary Search Tree M
- ☐ 450 Delete Node in a BST M
- ☐ 700 Search in a Binary Search Tree E

PATTERN 18

Backtracking Basics

6 MED

Choose → explore → un-choose; the core recursion template.

- ☐ 46 Permutations M
- ☐ 47 Permutations II M
- ☐ 77 Combinations M
- ☐ 78 Subsets M
- ☐ 90 Subsets II M
- ☐ 39 Combination Sum M

PATTERN 19

Backtracking with Constraints

4 MED 2 HARD

Prune early with feasibility checks to tame exponential blowup.

- ☐ 40 Combination Sum II M
- ☐ 17 Letter Combinations of a Phone Number M

PATTERN 20

Graph BFS / DFS

1 EASY 5 MED

Grid and adjacency traversal; mark visited to avoid loops.

- ☐ 200 Number of Islands M
- ☐ 695 Max Area of Island M

- ☐ 79 Word Search M
- ☐ 131 Palindrome Partitioning M
- ☐ 51 N-Queens H
- ☐ 52 N-Queens II H

- ☐ 733 Flood Fill E
- ☐ 994 Rotting Oranges M
- ☐ 1091 Shortest Path in Binary Matrix M
- ☐ 1254 Number of Closed Islands M

PATTERN 21

Topological Sort / DAG

5 MED 1 HARD

Order by dependencies; detect cycles via in-degree or coloring.

- ☐ 207 Course Schedule M
- ☐ 210 Course Schedule II M
- ☐ 802 Find Eventual Safe States M
- ☐ 1462 Course Schedule IV M
- ☐ 1203 Sort Items by Groups Respecting Dependencies H
- ☐ 2115 Find All Possible Recipes from Given Supplies M

PATTERN 22

Union Find / DSU

5 MED 1 HARD

Near- $O(1)$ connectivity with path compression + union by rank.

- ☐ 547 Number of Provinces M
- ☐ 684 Redundant Connection M
- ☐ 1319 Number of Operations to Make Network Connected M
- ☐ 1579 Remove Max Number of Edges to Keep Graph Fully Traversable H
- ☐ 990 Satisfiability of Equality Equations M
- ☐ 1202 Smallest String With Swaps M

PATTERN 23

Shortest Path

6 MED

Dijkstra, Bellman-Ford, or BFS depending on edge weights.

- ☐ 743 Network Delay Time M
- ☐ 787 Cheapest Flights Within K Stops M
- ☐ 1514 Path with Maximum Probability M

PATTERN 24

MST / Graph Greedy

3 MED 3 HARD

Kruskal/Prim to connect all nodes at minimum cost.

- ☐ 1584 Min Cost to Connect All Points M
- ☐ 1135 Connecting Cities With Minimum Cost M
- ☐ 1168 Optimize Water Distribution in a Village H

- ☐ 1631 Path With Minimum Effort M
- ☐ 1334 Find the City With the Smallest Number of Neighbors at a Threshold Distance M
- ☐ 1976 Number of Ways to Arrive at Destination M

- ☐ 1489 Find Critical and Pseudo-Critical Edges in Minimum Spanning Tree H
- ☐ 778 Swim in Rising Water H
- ☐ 1102 Path With Maximum Minimum Value M

PATTERN 25

Trie

5 MED 1 HARD

Prefix tree for fast word insertion, search, and autocomplete.

- ☐ 208 Implement Trie (Prefix Tree) M
- ☐ 211 Design Add and Search Words Data Structure M
- ☐ 212 Word Search II H
- ☐ 648 Replace Words M
- ☐ 677 Map Sum Pairs M
- ☐ 1268 Search Suggestions System M

PATTERN 26

Bit Manipulation

5 EASY 1 MED

XOR tricks, bit counting, and masks for $O(1)$ set operations.

- ☐ 136 Single Number E
- ☐ 137 Single Number II M
- ☐ 191 Number of 1 Bits E
- ☐ 338 Counting Bits E
- ☐ 268 Missing Number E
- ☐ 190 Reverse Bits E

PATTERN 27

1D DP Basics

1 EASY 5 MED

Define state over one index; build the table bottom-up.

- ☐ 70 Climbing Stairs E
- ☐ 198 House Robber M
- ☐ 213 House Robber II M
- ☐ 322 Coin Change M
- ☐ 279 Perfect Squares M

PATTERN 28

Kadane / Max Subarray

6 MED NEW

Running best-ending-here; the DP that feels greedy.

- ☐ 53 Maximum Subarray M
- ☐ 918 Maximum Sum Circular Subarray M
- ☐ 152 Maximum Product Subarray M
- ☐ 978 Longest Turbulent Subarray M

☐ 300

Longest Increasing Subsequence

M

☐ 413

Arithmetic Slices

M

☐ 300

Longest Increasing Subsequence

M

PATTERN 29

2D Grid DP

6 MED

NEW

Fill a matrix where each cell depends on neighbors above/left.

☐ 62

Unique Paths

M

☐ 64

Minimum Path Sum

M

☐ 120

Triangle

M

☐ 931

Minimum Falling Path Sum

M

☐ 221

Maximal Square

M

☐ 1277

Count Square Submatrices with All Ones

M

PATTERN 30

Knapsack / Subset DP

6 MED

NEW

Include-or-exclude decisions over a capacity or target.

☐ 416

Partition Equal Subset Sum

M

☐ 494

Target Sum

M

☐ 474

Ones and Zeroes

M

☐ 518

Coin Change II

M

☐ 377

Combination Sum IV

M

☐ 139

Word Break

M

PATTERN 31

String / Sequence DP

5 MED

1 HARD

NEW

2D table over two strings; edit, match, and subsequence.

☐ 1143

Longest Common Subsequence

M

☐ 72

Edit Distance

M

☐ 516

Longest Palindromic Subsequence

M

☐ 10

Regular Expression Matching

H

☐ 97

Interleaving String

M

☐ 91

Decode Ways

M

PATTERN 32

Palindrome Patterns

3 EASY

3 MED

NEW

Expand-around-center or interval DP for palindromic structure.

☐ 5

Longest Palindromic Substring

M

☐ 647

Palindromic Substrings

M

☐ 9

Palindrome Number

E

☐ 125

Valid Palindrome

E

☐ 680

Valid Palindrome II

E

☐ 516

Longest Palindromic Subsequence

M

PATTERN 33

Matrix Manipulation

6 MED

NEW

In-place rotation, spiral traversal, and marker tricks.

- ☐ 48 Rotate Image M
- ☐ 54 Spiral Matrix M
- ☐ 73 Set Matrix Zeroes M
- ☐ 289 Game of Life M
- ☐ 36 Valid Sudoku M
- ☐ 238 Product of Array Except Self M

PATTERN 34

Cyclic Sort / Index-as-Hash

3 EASY

2 MED

1 HARD

NEW

Use values as indices to find missing/duplicate in $O(1)$ space.

- ☐ 287 Find the Duplicate Number M
- ☐ 41 First Missing Positive H
- ☐ 448 Find All Numbers Disappeared in an Array E
- ☐ 442 Find All Duplicates in an Array M
- ☐ 268 Missing Number E
- ☐ 169 Majority Element E

PATTERN 35

Design / Data Structures

1 EASY

3 MED

2 HARD

NEW

Compose hash + list + heap to hit strict time bounds.

- ☐ 146 LRU Cache M
- ☐ 460 LFU Cache H
- ☐ 155 Min Stack M
- ☐ 380 Insert Delete GetRandom $O(1)$ M
- ☐ 295 Find Median from Data Stream H
- ☐ 225 Implement Stack using Queues E

PATTERN 36

Boyer-Moore / Voting

2 EASY

3 MED

1 HARD

NEW

Cancel out pairs to find the majority in $O(1)$ space.

- ☐ 169 Majority Element E
- ☐ 229 Majority Element II M
- ☐ 287 Find the Duplicate Number M
- ☐ 41 First Missing Positive H
- ☐ 448 Find All Numbers Disappeared in an Array E
- ☐ 442 Find All Duplicates in an Array M

PATTERN 37

PATTERN 38

Expand & Count DP

1 EASY

5 MED

NEW

Combinatorial DP: decode, delete-and-earn, arithmetic slices.

- ☐ 91 Decode Ways M
- ☐ 740 Delete and Earn M
- ☐ 413 Arithmetic Slices M
- ☐ 377 Combination Sum IV M
- ☐ 279 Perfect Squares M
- ☐ 70 Climbing Stairs E

Reservoir / Randomized

1 EASY

4 MED

1 HARD

NEW

O(1)-extra sampling and randomized selection structures.

- ☐ 380 Insert Delete GetRandom O(1) M
- ☐ 295 Find Median from Data Stream H
- ☐ 146 LRU Cache M
- ☐ 215 Kth Largest Element in an Array M
- ☐ 973 K Closest Points to Origin M
- ☐ 703 Kth Largest Element in a Stream E

PATTERN 39

Prefix-Product & Rolling

6 MED

NEW

Products around each index and rolling-hash style scans.

- ☐ 238 Product of Array Except Self M
- ☐ 152 Maximum Product Subarray M
- ☐ 918 Maximum Sum Circular Subarray M
- ☐ 53 Maximum Subarray M
- ☐ 560 Subarray Sum Equals K M
- ☐ 974 Subarray Sums Divisible by K M

PATTERN 40

In-place Linked List DP-ish

1 EASY

4 MED

1 HARD

NEW

Reuse node pointers to solve in O(1) extra space.

- ☐ 19 Remove Nth Node From End of List M
- ☐ 24 Swap Nodes in Pairs M
- ☐ 25 Reverse Nodes in k-Group H
- ☐ 92 Reverse Linked List II M
- ☐ 138 Copy List with Random Pointer M
- ☐ 234 Palindrome Linked List E

